Universe is made almost entirely of matter. The reason for this engages the deepest cosmological issues, which need not detain us here. But if there was only a one-particle-in-a-billion difference in the preponderance of matter over anti-matter at the beginning, even this would be enough to explain the Universe we see today.

Williamson imagined that humans in the twenty-second century would move asteroids around by the controlled mutual annihilation of matter and anti-matter. All the resulting gamma rays, if collimated, would make a potent rocket exhaust. The anti-matter would be available in the main asteroid belt (between the orbits of Mars and Jupiter), because this was his explanation for the *existence* of the asteroid belt. In the remote past, he proposed, an intruder anti-matter worldlet arrived in the Solar System from the depths of space, impacted, and annihilated what was then an Earthlike planet, fifth from the Sun. The fragments of this mighty collision are the asteroids, and some of them are still made of anti-matter. Harness an anti-asteroid—Williamson recognized that this might be tricky—and you can move worlds around at will.

At the time, Williamson's ideas were futuristic, but far from foolish. Some of "Collision Orbit" can be considered visionary. Today, however, we have good reason to believe that there are no significant amounts of anti-matter in the Solar System, and that the asteroid belt, far from being a fragmented terrestrial planet, is an enormous array of small bodies prevented (by the gravitational tides of Jupiter) from forming an Earthlike world.

However, we do generate (very) small amounts of anti-matter in nuclear accelerators today, and we will probably be able to manufacture much larger amounts by the twenty-second century. Because it is so efficient—converting *all* of the matter into energy, $E = mc^2$, with 100 percent efficiency—perhaps anti-matter engines will be a practical technology by then, vindicating Williamson. Failing that, what energy sources can we realistically expect to be available, to reconfigure asteroids, to light them, warm them, and move them around?

The Sun shines by jamming protons together and turning them into helium nuclei. Energy is released in the process, although with